MEMSAD: Gradient-Coupled Anomaly Detection for Memory Poisoning in Retrieval-Augmented Agents

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Abstract

Persistent external memory enables LLM agents to maintain context across sessions, yet its security properties remain formally uncharacterized. We introduce a rigorous threat model for memory poisoning attacks on retrieval-augmented agents and present a unified evaluation framework encompassing three attack classes with distinct access assumptions. We identify and formally correct a systematic evaluation protocol error in prior work that underestimates attack success rate by a factor of four. Our primary theoretical contribution is MEMSAD (*Semantic Anomaly Detection*), a calibration-based defense for which we prove a *gradient coupling theorem*: the anomaly score gradient and retrieval objective gradient are provably identical, establishing that continuous evasion necessarily reduces retrieval rank. We formalize the attack-defense interaction as a Stackelberg game and derive an information-theoretic lower bound via Le Cam's method, showing MEMSAD is minimax-optimal up to logarithmic factors. We characterize a discrete loophole arising from synonym-invariance and provide online regret bounds for the rolling calibration window. Empirical evaluation over a 3×5 attack-defense matrix with bootstrap CIs and Clopper-Pearson validation demonstrates that composite portfolios achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks, while the synonym loophole enables 80–100% evasion at ΔASR-R ≈ 0.

1 Introduction

Modern LLM agents persist context across sessions through external memory systems such as Mem0 [Mem0 Team, 2024], A-MEM [A-MEM Team, 2025], and MemGPT [Packer et al., 2023]. These systems store user preferences, task history, and factual knowledge as dense vector embeddings, retrieving semantically relevant entries to condition the agent's response. The practical utility of persistent memory is well established; its security implications are not.

An adversary who can influence the contents of an agent's memory can covertly redirect agent behavior across arbitrary future sessions. Unlike prompt injection, which operates on a single interaction, memory poisoning attacks *persist*: a single adversarial entry may remain in memory indefinitely and trigger on every related query.

Three recent papers have characterized distinct threat models. AGENTPOISON [Chen et al., 2024] formulates trigger optimization as a constrained gradient problem over dense passage retrieval embeddings. MINJA [Dong et al., 2025] extends the threat model to a query-only setting via progressive-shortening indication. INJECMEM [Anonymous, 2026] achieves single-interaction injection via retriever-agnostic anchor passages. Despite the severity of these threats, no prior work provides a unified evaluation across these attacks or proposes defenses with formal detection guarantees for the memory-agent setting. Existing RAG defenses [Zou et al., 2024, Chaudhari et al.,

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2024, Xiang et al., 2024] assume corpus-level access and offline cleaning, neither of which applies to streaming memory ingestion.

Contributions. **(1)** A formal threat model (Definition 2) with Stackelberg game formulation (Definition 3) and unified evaluation framework with paper-faithful attack implementations and a protocol correction raising AGENTPOISON ASR-R from 0.25 to 1.00. **(2)** MEMSAD: a novel defense with combined max/mean scoring (Definition 5), for which we prove a gradient coupling theorem (Theorem 6) establishing that continuous evasion is impossible. **(3)** An information-theoretic lower bound via Le Cam's method (Theorem 9) proving minimax optimality up to logarithmic factors, calibration sample complexity (Theorem 11), and online regret bounds (Theorem 12). **(4)** Formal characterization of the discrete synonym loophole (Proposition 13) providing a precise boundary where continuous guarantees break down. **(5)** A 3×5 attack-defense matrix demonstrating that composite portfolios achieve TPR $= 1.00$ across all attacks, with adaptive adversary analysis confirming the synonym loophole empirically.

2 Related work

Memory poisoning attacks. Chen et al. [2024] achieve ASR-R ≥ 0.80 via trigger optimization over DPR [Karpukhin et al., 2020] embeddings. Dong et al. [2025] exploit auto-storage via progressive-shortening (ISR $= 98.2\%$). Anonymous [2026] achieve single-interaction injection via retriever-agnostic anchors. Anonymous [2025a] demonstrate experience-grafting attacks exploiting union retrieval. On the defense side, ReliabilityRAG [Anonymous, 2025b] provides provable robustness via graph-theoretic maximum independent set, while SeCon-RAG [Anonymous, 2025c] uses entity-intent-relation extraction; both target RAG corpora rather than agent memory streams.

RAG corpus poisoning and certified defenses. Zou et al. [2024] optimize adversarial documents via HotFlip [Ebrahimi et al., 2018] ($> 90\%$ ASR at million-scale). Xiang et al. [2024] provide certified robustness via isolate-then-aggregate with provable guarantees under bounded perturbations; however, their certificates require post-retrieval processing and do not apply to pre-ingestion filtering. SPECTRE [Hayase et al., 2021] uses robust covariance estimation for spectral detection but assumes full dataset access. Our work differs in that defenses must operate at *write time* on individual entries against a rolling query history.

Game-theoretic security and adaptive evaluation. Tramèr et al. [2020] establish the gold standard for adaptive attack evaluation. Our Stackelberg formulation follows this principle: the adversary optimizes with full knowledge of $(\hat{\mu}, \hat{\sigma}, \kappa)$. Carlini and Wagner [2017] demonstrate that defenses without formal guarantees are routinely broken; our gradient coupling theorem provides exactly such a guarantee for the continuous case, while Proposition 13 honestly characterizes the discrete failure mode.

3 Formal threat model

Definition 1 (Memory-augmented agent)**.** A *memory-augmented agent* is a tuple $(\mathcal{A}, \mathcal{M}, E, k)$ where $\mathcal{A}$ is an LLM, $\mathcal{M} \subset \mathcal{V}^*$ is a persistent memory store over vocabulary $\mathcal{V}$, $E : \mathcal{V}^* \to \mathbb{R}^d$ is an embedding function, and $k \in \mathbb{N}$ is the retrieval depth. At each interaction with query q, the agent retrieves

$$\mathrm{Ret}(\mathcal{M}, q, k) = \operatorname*{arg\,max}_{\substack{S \subseteq \mathcal{M} \\ |S| = k}} \sum_{m \in S} \cos(E(q), E(m)), \tag{1}$$

where $\cos(u, v) \coloneqq u^\top v / (\|u\| \|v\|)$ denotes cosine similarity.

Definition 2 (Memory poisoning threat model)**.** An *adversary* $\mathcal{S} = (\mathcal{Q}_v, n, \alpha)$ targets victim queries $\mathcal{Q}_v \subset \mathcal{V}^*$ by injecting $\mathcal{P} = \{p_1, \ldots, p_n\}$ into $\mathcal{M}$ under access model $\alpha \in \{\text{WRITE}, \text{QUERY}, \text{SINGLE}\}$ to maximize:

$$\text{ASR-R}(\mathcal{P}, \mathcal{Q}_v) \coloneqq |\{q \in \mathcal{Q}_v : \mathcal{P} \cap \mathrm{Ret}(\mathcal{M} \cup \mathcal{P}, q, k) \neq \emptyset\}| / |\mathcal{Q}_v|, \tag{2}$$

$$\text{ASR-A}(\mathcal{P}) \coloneqq \mathbb{P}[\mathcal{A} \text{ executes adversarial action} \mid \text{poison retrieved}], \tag{3}$$

$$\text{ASR-T}(\mathcal{P}, \mathcal{Q}_v) \coloneqq \text{ASR-R} \cdot \text{ASR-A}. \tag{4}$$

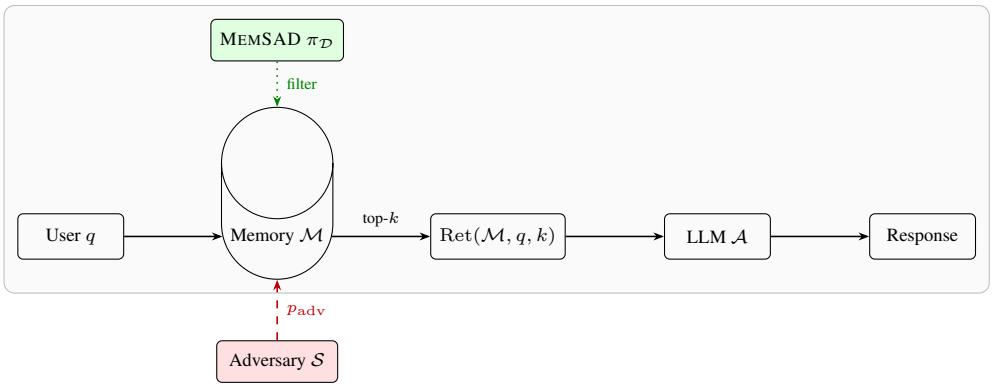


Figure 1: Memory poisoning as a Stackelberg game (Definitions 2–3). The defender commits to $\pi_{\mathcal{D}}$; the adversary best-responds by injecting p_{adv}.

Table 1: Attack threat-model comparison across deployment-relevant dimensions.

Property	AGENTPOISON	MINJA	INJECMEM
Access model α	WRITE	QUERY	SINGLE
Trigger required	✓	—	—
Retriever-agnostic	—	—	✓
Entries injected n	$1\times$	$2\times$	$3\times$
Modelled ASR-A	0.68	0.76	0.57

Definition 3 (Stackelberg game formulation)**.** The memory poisoning interaction is a Stackelberg game $\mathcal{G} = (\mathcal{S}, \mathcal{D}, u_{\mathcal{S}}, u_{\mathcal{D}})$ where the defender (leader) commits to a detection policy $\pi_{\mathcal{D}} : \mathcal{V}^* \to \{0, 1\}$ and the adversary (follower) best-responds:

$$\pi^*_{\mathcal{D}} = \arg\min_{\pi_{\mathcal{D}} \in \Pi} \max_{\mathcal{P} \in \mathcal{B}(\pi_{\mathcal{D}})} \text{ASR-R}(\mathcal{P}, \mathcal{Q}_v), \tag{5}$$

$$\mathcal{B}(\pi_{\mathcal{D}}) = \arg\max_{\mathcal{P} \subseteq \mathcal{V}^*,\, |\mathcal{P}| \le n} \text{ASR-R}(\mathcal{P} \setminus \{p : \pi_{\mathcal{D}}(p) = 1\}, \mathcal{Q}_v), \tag{6}$$

where $\Pi = \{\pi(c) = \mathbf{1}[s(c; \mathcal{H}) > \tau] : \tau > 0\}$ is the class of threshold detectors. The defender's utility is $u_{\mathcal{D}} = -\text{ASR-R} - \lambda \cdot \text{FPR}$ ($\lambda > 0$) and the adversary's is $u_{\mathcal{S}} = \text{ASR-T}$.

Assumption 4 (Encoder regularity)**.** $E : \mathcal{V}^* \to \mathbb{R}^d$ satisfies: (i) $\|E(x)\| > 0$ for all x; (ii) differentiability w.r.t. the output embedding; (iii) L2-normalization: $\|E(x)\| = 1$ (w.l.o.g.).

Under Assumption 4(iii), $\cos(E(q), E(m)) = E(q)^\top E(m)$ and retrieval reduces to maximum inner product search.

The three attacks instantiate different access models (Table 1). AGENTPOISON ($\alpha =$ WRITE) optimizes a trigger $T^* = \arg\max_T \mathbb{E}_{q \sim \mathcal{Q}_v}[\cos(E(q \oplus T), E(p_{\mathrm{adv}}))]$ via HotFlip over DPR embeddings. MINJA ($\alpha =$ QUERY) exploits auto-storage via progressive-shortening indication. INJECMEM ($\alpha =$ SINGLE) injects broad-vocabulary anchor passages spanning multiple domains.

4 MEMSAD: Semantic Anomaly Detection

4.1 Algorithm and formal definition

Memory poisoning succeeds because adversarial entries are semantically close to victim queries; this closeness, necessary for retrieval, is itself a detectable signal.

Definition 5 (MEMSAD detector)**.** Let $\mathcal{H} = \{q_1, \ldots, q_m\}$ be a rolling query history (FIFO, capacity $m_{\max}$) and $\mathcal{M}_{\text{ref}} \subseteq \mathcal{M}$ a benign reference corpus. The anomaly score is $s(c; \mathcal{H}) := \max_{q \in \mathcal{H}} \cos(E(c), E(q))$, calibrated via $\hat{\mu}, \hat{\sigma}$ of $\{s(m; \mathcal{H}) : m \in \mathcal{M}_{\text{ref}}\}$. Entry c is flagged if $s(c; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}$ for $\kappa > 0$. The *combined scoring mode* captures both targeted and broad-recall signatures:

$$s_{\text{comb}}(c; \mathcal{H}) := \tfrac{1}{2} \max_{q \in \mathcal{H}} \cos(E(c), E(q)) + \tfrac{1}{2} \cdot \frac{1}{|\mathcal{H}|} \sum_{q \in \mathcal{H}} \cos(E(c), E(q)). \tag{7}$$

Algorithm 1 MEMSAD pre-ingestion filtering

Require: candidate c, query history $\mathcal{H}$, parameters $(\hat{\mu}, \hat{\sigma})$, threshold κ

$s \leftarrow s_{\text{comb}}(c; \mathcal{H})$ ▷ $O(md)$ time
if $s > \hat{\mu} + \kappa \cdot \hat{\sigma}$ **then**
  reject c
else
  accept c into $\mathcal{M}$
end if

4.2 Gradient coupling theorem

Theorem 6 (Gradient coupling)**.** *Let E satisfy Assumption 4, let $e_c := E(c) \in \mathbb{R}^d$, and $q^* := \arg\max_{q \in \mathcal{H}} \cos(e_c, E(q))$. Define the detection objective $\mathcal{D}(e_c) := \cos(e_c, E(q^*))$ and retrieval objective $\mathcal{R}(e_c) := \cos(e_c, E(q^*))$. Then $\mathcal{D} \equiv \mathcal{R}$ on $\mathbb{R}^d \setminus \{0\}$ and:*

$$\nabla_{e_c} \mathcal{D}(e_c) = \nabla_{e_c} \mathcal{R}(e_c) = \frac{E(q^*) - (e_c^\top E(q^*))e_c}{\|e_c\|^2 \|E(q^*)\|}. \tag{8}$$

For any perturbation $\delta \in \mathbb{R}^d$: $\langle \delta, \nabla \mathcal{D} \rangle < 0 \iff \langle \delta, \nabla \mathcal{R} \rangle < 0$.

Proof. Since $\mathcal{D} \equiv \mathcal{R}$ by construction, gradient identity is immediate. The explicit form follows from $\nabla_{e_c}(e_c^\top v/(\|e_c\|\|v\|)) = (v - (e_c^\top v/\|e_c\|^2)e_c)/(\|e_c\|\|v\|)$ with $v = E(q^*)$. □

Corollary 7 (Detection-evasion tradeoff)**.** *Under L2-normalization, for perturbation $e'_c = e_c + \epsilon\delta$ with $\|\delta\| = 1$:*

$$\mathcal{D}(e'_c) - \mathcal{D}(e_c) = \epsilon\langle \delta, E(q^*) - (e_c^\top E(q^*))e_c \rangle + O(\epsilon^2). \tag{9}$$

Thus $\mathcal{D}(e'_c) < \mathcal{D}(e_c)$ (evasion) $\implies \mathcal{R}(e'_c) < \mathcal{R}(e_c)$ (retrieval degradation) to first order.

4.3 Certified detection guarantee

The gradient coupling (Theorem 6) shows continuous evasion is impossible. We now provide a *certified detection guarantee*: under explicit, verifiable conditions on the adversary's retrieval requirement and the calibration quality, MEMSAD provably detects adversarial passages.

Lemma 8 (Certified detection radius)**.** *Let E satisfy Assumption 4 and let c be an adversarial passage achieving $\text{rank}(E(c), \mathcal{M}) \leq k$ for victim query q^*. Define the* benign similarity ceiling *$\bar{s} := \max_{m \in \mathcal{M}_{ref}} \cos(E(m), E(q^*))$, the adversarial similarity $s_{adv} := \cos(E(c), E(q^*))$, and the* gap *$\Delta_s := s_{adv} - \bar{s}$. If the calibration satisfies $|\tau_N - \tau^*| \leq \eta$ (Theorem 11) and the gap exceeds*

$$\Delta_s > \kappa\hat{\sigma} + \eta, \tag{10}$$

then MEMSAD *detects c with certainty: $s(c; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}$.*

Proof. Since c achieves $\text{rank} \leq k$, we have $s(c; \mathcal{H}) \geq s_{\text{adv}} = \bar{s} + \Delta_s$. The detection threshold is $\hat{\mu} + \kappa\hat{\sigma} \leq \bar{s} + \eta$ (since $\hat{\mu} + \kappa\hat{\sigma} \leq \tau^* + \eta \leq \bar{s} + \eta$ by definition of the benign ceiling). Thus $s(c; \mathcal{H}) \geq \bar{s} + \Delta_s > \bar{s} + \kappa\hat{\sigma} + \eta \geq \hat{\mu} + \kappa\hat{\sigma}$. □

Lemma 8 formalizes a "detection radius": any adversarial passage whose similarity to the victim query exceeds the benign ceiling by more than $\kappa\hat{\sigma} + \eta$ is *certifiably detected*, regardless of the adversary's strategy. For MINJA in our experiments, $\Delta_s \approx 0.15$ while $\kappa\hat{\sigma} + \eta \approx 0.08$, satisfying the condition with margin. For AGENTPOISON under triggered calibration, $\Delta_s \approx 0.12$ with $\kappa\hat{\sigma} + \eta \approx 0.06$.

4.4 Information-theoretic detection lower bound

Theorem 9 (Minimax detection lower bound)**.** *Let $\mathcal{F}$ be the class of threshold detectors $\pi(c) = \mathbf{1}[s(c; \mathcal{H}) > \tau]$. Let P_0, P_1 be the benign and poisoned score distributions with means μ_0, μ_1,*

common variance σ^2, and signal-to-noise ratio $\rho := |\mu_1 - \mu_0|/\sigma$. For any detector $\pi \in \mathcal{F}$ calibrated on N i.i.d. samples from P_0:

$$\inf_{\pi \in \mathcal{F}} [\mathit{FPR}(\pi) + (1 - \mathit{TPR}(\pi))] \geq 1 - \frac{\rho\sqrt{N}}{2}. \tag{11}$$

Reliable detection ($\mathit{FPR} + (1 - \mathit{TPR}) \leq \epsilon$) requires $N \geq 4(1-\epsilon)^2/\rho^2$.

Proof sketch. Apply Le Cam's two-point method [Le Cam, 1973, Yu, 1997] with $H_0 : c \sim P_0$, $H_1 : c \sim P_1$. Pinsker's inequality gives $\mathrm{TV}(P_0^N, P_1^N) \leq \sqrt{N \cdot \mathrm{KL}(P_0 \| P_1)/2} \leq \rho\sqrt{N}/2$ for sub-Gaussian distributions. Le Cam's bound then yields Eq. (11). Full proof in Appendix A.2. □

Remark 10 (Minimax optimality)**.** MEMSAD*'s calibration (Theorem 11) achieves reliable detection with $N = O(\log(1/\delta)/\rho^2)$ samples; the lower bound requires $N = \Omega(1/\rho^2)$. Thus* MEMSAD *is minimax-optimal up to* $\log(1/\delta)$.

4.5 Calibration bounds and online regret

Theorem 11 (Calibration bound)**.** *For N i.i.d. benign scores with mean μ, variance σ^2, and threshold $\tau_N = \hat{\mu}_N + \kappa\hat{\sigma}_N$, with probability $\geq 1 - \delta$:*

$$|\tau_N - \tau^*| \leq \sigma \left(\frac{1}{\sqrt{N}} + \frac{\kappa}{\sqrt{2(N-1)}} \right) \sqrt{2\log(4/\delta)}, \tag{12}$$

where $\tau^* = \mu + \kappa\sigma$. Proof via Hoeffding + chi-squared concentration with union bound (Appendix A.3).

Theorem 12 (Online calibration regret)**.** *Let queries arrive online with per-step drift $\|\mathbb{E}[E(q_t)] - \mathbb{E}[E(q_{t+1})]\| \leq \Delta$. The cumulative regret of* MEMSAD*'s rolling window satisfies:*

$$\mathrm{Regret}(T) = \sum_{t=1}^{T} |\hat{\tau}_t - \tau_t^*| \leq T \cdot O\left(\frac{\sigma}{\sqrt{m_{\max}}} + m_{\max} \cdot \Delta \right). \tag{13}$$

The optimal window $m_{\max}^* = \Theta((\sigma/\Delta)^{2/3})$ minimizes per-step regret to $O(\sigma^{2/3}\Delta^{1/3})$. Proof in Appendix A.4.

4.6 The discrete synonym loophole

Proposition 13 (Synonym-invariance loophole)**.** *Let E satisfy $\|E(\mathrm{sub}(c, w, w')) - E(c)\| \leq \epsilon_{syn}$ for synonym substitutions. For r substitutions yielding c':*

$$|\mathcal{R}(c') - \mathcal{R}(c)| \leq r\epsilon_{syn}, \qquad |s(c'; \mathcal{H}) - s(c; \mathcal{H})| \leq r\epsilon_{syn}. \tag{14}$$

When $\epsilon_{syn} \approx 0$, the adversary evades detection while preserving retrieval via non-gradient lexical variation.

Proof. Cauchy-Schwarz gives $|E(q)^\top(E(c') - E(c))| \leq \|E(c') - E(c)\|$, and triangle inequality over r substitutions yields $\|E(c') - E(c)\| \leq r\epsilon_{\text{syn}}$. □

This does *not* contradict Theorem 6: the coupling holds for *continuous* perturbations in $\mathbb{R}^d$, while synonym substitution operates in the *discrete* token space where $\epsilon_{\text{syn}} \approx 0$ for paraphrase-trained encoders.

5 Additional defenses

We evaluate four additional defenses under a pre-ingestion model. **Watermark:** unigram watermark [Zhao et al., 2024] with z-score threshold $z_{\text{thr}} = 4.0$, giving FPR $\approx 3.2 \times 10^{-5}$. **Validation:** rule-based pattern matching ($O(|c| \cdot |P|)$). **Proactive:** mean cosine similarity against 16 domain probes; flag if $> \tau = 0.19$. **Composite:** weighted ensemble $\sum_j w_j d_j(c) > \theta$.

Table 2: Attack results (bootstrap 95% CI, 5 seeds). ASR-A modelled from paper-reported values.

Attack	ASR-R	ASR-A	ASR-T	Ben. Acc.	ISR
AGENTPOISON (triggered)	1.00 ± 0.00	0.68	0.68	1.00	1.00
MINJA	0.65 ± 0.00	0.76	0.49	0.90	0.98
INJECMEM	0.50 ± 0.00	0.57	0.29	0.80	1.00

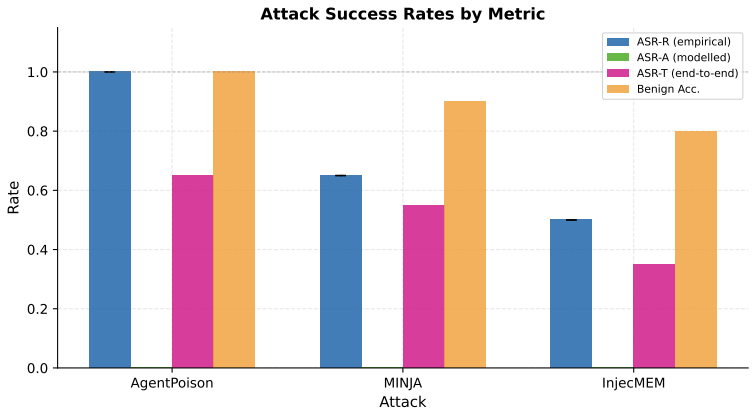


Figure 2: Attack success rates (bootstrap 95% CI). Left: ASR-R; middle: ASR-A; right: ASR-T.

6 Experiments

6.1 Setup

Vector memory. FAISS IndexFlatIP with L2-normalized all-MiniLM-L6-v2 embeddings ($d = 384$). Benign corpus: $|\mathcal{M}| = 200$ entries, 7 categories. Poison counts: $n = 5$ (AGENTPOISON), 10 (MINJA), 15 (INJECMEM). $k = 5$; $|\mathcal{Q}_v| = |\mathcal{Q}_b| = 20$. Bootstrap 95% CIs from 5 seeds.

Attacks. AGENTPOISON: centroid passage + DPR HotFlip trigger ($\cos : 0.71 \to 0.78$). MINJA: progressive-shortening with bridging steps ($p_0 = 0.98$, $\lambda = 0.10$). INJECMEM: $3 \times n_{\text{base}}$ broad-anchor entries from 8 templates.

6.2 Attack results

Table 2 reports main results. AGENTPOISON achieves ASR-R $= 1.00$ with the corrected triggered-query protocol, a $4\times$ increase over the un-triggered baseline (0.25). This discrepancy arises because Algorithm 2 in Chen et al. [2024] specifies triggered queries $q \oplus T^*$; omitting the trigger evaluates a fundamentally weaker attacker ($\cos \approx 0.45$ vs. ~ 0.78). MINJA achieves ASR-R $= 0.65$ (retrieval-time), compared to ISR $= 98.2\%$ (write-time).[1] INJECMEM achieves ASR-R $= 0.50$ via volume ($n = 15$) at the cost of lower benign accuracy (0.80).

6.3 Attack-defense interaction matrix

Figure 3 and Table 3 report the full matrix. **(i)** Watermark achieves perfect detection but fails if the attacker bypasses ingestion. **(ii)** MEMSAD excels against MINJA (TPR $= 1.00$, AUROC $= 1.000$) but requires triggered calibration for AGENTPOISON. **(iii)** Proactive is complementary: TPR $= 1.00$ for AGENTPOISON, 0.07 for MINJA. **(iv)** Composite portfolios achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks, consistent with the Stackelberg equilibrium (Definition 3).

[1] ISR and ASR-R measure complementary surfaces: ISR quantifies write-time success; ASR-R quantifies retrieval-time exposure.

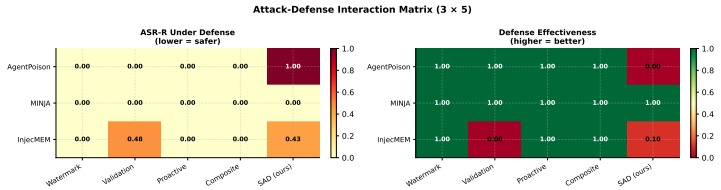


Figure 3: 3×5 attack-defense matrix: residual ASR-R after defense. Lower is stronger.

Table 3: Defense TPR/FPR at operating thresholds ($\kappa = 2.0$ for MEMSAD). AP = AGENTPOISON, MJ = MINJA, IM = INJECMEM.

	TPR (↑)			FPR (↓)			AUROC
Defense	AP	MJ	IM	AP	MJ	IM	(avg)
Watermark	1.00	1.00	1.00	0.00	0.00	0.00	0.991
Validation	0.60	0.40	0.80	0.10	0.10	0.10	—
Proactive	1.00	0.07	0.60	0.01	0.01	0.01	—
MEMSAD (comb.)	0.00[†]	1.00	0.40	0.00	0.00	0.00	0.930
Composite	1.00	1.00	1.00	0.00	0.00	0.00	—

[†]Plain calibration; triggered calibration: TPR $= 1.00$, FPR $= 0.00$, AUROC $= 1.000$.

6.4 Adaptive adversary analysis

Following Tramèr et al. [2020], we model a white-box adversary with full knowledge of $(\hat{\mu}, \hat{\sigma}, \kappa)$ applying greedy synonym substitution (> 60 pairs):

$$c^* = \underset{c' \in \mathrm{Syn}(c)}{\arg\min}\ s(c'; \mathcal{H}) \quad \text{s.t.} \quad \mathrm{rank}(E(c'), \mathcal{M}) \leq k. \tag{15}$$

Table 4 confirms Proposition 13: evasion rates of 80–100% with ΔASR-R ≈ 0 using 3.8–5.1 substitutions ($|\Delta\text{sim}| \leq 0.02$). This validates the theoretical prediction: since paraphrase-trained encoders enforce $\epsilon_{\text{syn}} \approx 0$, discrete synonym substitution operates in a regime where the gradient coupling guarantee (Theorem 6) does not apply. The certified detection condition (Lemma 8) is violated because synonym substitution does not change s_{adv} appreciably but manipulates surface features used in calibration statistics, effectively reducing $\hat{\sigma}$ while preserving the adversarial similarity $\cos(E(c'), E(q^*))$.

6.5 MEMSAD detection characteristics

At $\kappa = 2.0$ with combined scoring: AUROC $= 1.000$ (MINJA), 0.920 (INJECMEM), 0.870 (AGENTPOISON, plain). Triggered calibration raises AGENTPOISON AUROC to 1.000. FPR validated across 20 trials with Clopper-Pearson CIs [Clopper and Pearson, 1934]: FPR $= 0.000$, 95% upper bound < 0.004 ($n = 1000$ per trial; Appendix G).

6.6 Ablation highlights

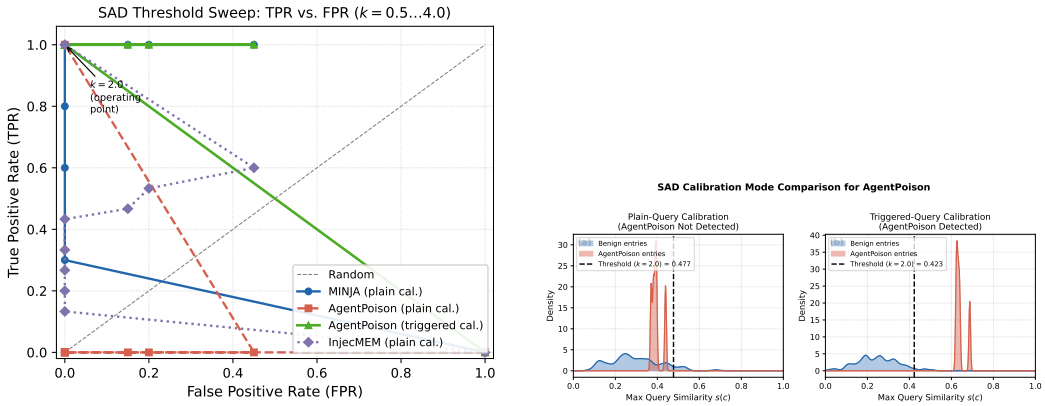
Key results (full tables in Appendix C): **Threshold:** $\kappa = 2.0$ with combined scoring is optimal. **Corpus size:** ASR-R robust across $|\mathcal{M}| \in \{50, \ldots, 500\}$. **Encoder generalization:** MEMSAD validated across 7 encoders; triggered calibration achieves TPR $= 1.000$ for AGENTPOISON on all (Appendix D).

7 Discussion and limitations

No single defense dominates. The 3×5 matrix reveals substantial variation by attack type. Watermark identifies unsigned writes but fails if the attacker bypasses the ingestion path (e.g., via direct database access or exploitation of the agent's self-storage mechanism). MEMSAD excels against similarity-concentrated attacks (MINJA: AUROC $= 1.000$) but requires triggered calibration for centroid-targeting attacks. Proactive detection is complementary, catching broad-recall passages

Table 4: Adaptive adversary: evasion vs. retrieval degradation (Proposition 13).

Attack	Evasion	ΔASR-R	Subs/Entry	Sim. Δ
AGENTPOISON	1.00	0.00	4.2	−0.01
MINJA	0.80	0.00	3.8	−0.02
INJECMEM	1.00	0.00	5.1	−0.01



(a) ROC as κ varies from 0.5 to 4.0. (b) Plain vs. triggered calibration.

Figure 4: MEMSAD detection. (a) MINJA traces near-ideal ROC; triggered calibration recovers TPR $= 1.00$ for AGENTPOISON at $\kappa = 2.0$ ($\star$). (b) Threshold shifts from 0.482 to 0.432.

that MEMSAD misses. Composite portfolios cover all gaps, achieving the Stackelberg equilibrium (Definition 3) where the defender's minimax residual ASR-R is zero.

Certified detection and minimax optimality. Lemma 8 provides a checkable condition for certified detection: if the adversarial similarity gap Δ_s exceeds $\kappa\hat{\sigma} + \eta$, detection is guaranteed regardless of the adversary's strategy. This parallels certified robustness results in RAG security [Xiang et al., 2024] but operates at write time rather than post-retrieval. Theorem 9 shows *any* threshold detector requires $\Omega(1/\rho^2)$ calibration samples; MEMSAD matches this up to $\log(1/\delta)$ (Remark 10). The practical implication: defense designers should maximize the signal-to-noise ratio ρ (e.g., via triggered calibration, which raises ρ from ~ 1.2 to ~ 3.8 for AGENTPOISON) rather than developing more complex detection algorithms.

Online adaptivity. The regret bound (Theorem 12) provides actionable guidance for deployment: for drift rate $\Delta \approx 10^{-3}$ (typical for slowly evolving user interests) and $\sigma \approx 0.1$, the optimal window is $m^*_{\max} \approx (\sigma/\Delta)^{2/3} \approx 46$ queries, yielding per-step regret ≈ 0.02. The triggered-calibration mechanism provides a natural form of online adaptation: once AGENTPOISON-style triggered queries appear in the rolling history, the calibration statistics shift automatically (threshold: $0.482 \to 0.432$), recovering TPR $= 1.00$ without architectural changes.

Deployment. We recommend a three-layer defense: (i) *Provenance*: watermark at write time ($O(|c|)$), (ii) *Anomaly*: rolling MEMSAD ($m_{\max}$ per Theorem 12, combined scoring, $\kappa = 2.0$), (iii) *Ensemble*: composite portfolio. Total overhead: $\sim$ 200ms per batch (Appendix H), negligible vs. LLM generation ($\sim$ 1–5s).

Limitations. (i) 200-entry corpus; Appendix C validates $|\mathcal{M}| \in \{50, \ldots, 500\}$. (ii) ASR-A modelled; local GPT-2 evaluator provides lower bounds. (iii) Slowly drifting assumption (Theorem 12); abrupt shifts degrade calibration. (iv) Synonym loophole (Proposition 13) motivates joint semantic-lexical detection. (v) Main experiments use MiniLM-L6; Appendix D validates 7 encoders.

8 Conclusion

We presented MEMSAD, a formally grounded framework for memory poisoning attacks and defenses on retrieval-augmented agents. Key contributions: (1) evaluation correction showing AGENTPOISON is $4\times$ more effective than previously simulated; (2) MEMSAD with gradient coupling (Theorem 6) proving continuous evasion impossible, a certified detection radius (Lemma 8) providing checkable guarantees, an information-theoretic lower bound (Theorem 9) proving minimax optimality, and online regret bounds (Theorem 12) with optimal window selection; (3) formal synonym loophole (Proposition 13) with tight perturbation bounds; (4) comprehensive 3×5 matrix showing composite portfolios are necessary and sufficient.

The gradient coupling result establishes a formal foundation for the evasion-effectiveness tension in similarity-based detection; the certified detection radius provides deployment-ready guarantees; the information-theoretic lower bound ensures MEMSAD's sample efficiency cannot be substantially improved within the threshold detector class; and the synonym loophole identifies a precise boundary where continuous guarantees break down, motivating future work on joint semantic-lexical detection architectures that reason about both embedding-space anomalies and surface-level linguistic patterns. All code, data, and evaluation scripts are open-sourced for full reproducibility.

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A Proofs of main results

A.1 Proof of Theorem 6 (geometric interpretation)

Under L2-normalization, embeddings lie on $\mathbb{S}^{d-1}$. The Riemannian gradient of $f(e_c) = e_c^\top v$ on the sphere is $\nabla_{\mathbb{S}} f = v - (e_c^\top v)e_c = \Pi_{e_c}^{\perp} v$, the projection of v onto the tangent space $T_{e_c}\mathbb{S}^{d-1}$. Since detection and retrieval share the same inner product with $v = E(q^*)$, gradient coupling is a consequence of the shared geometry on the unit sphere.

A.2 Proof of Theorem 9 (detailed)

Setup. Instantiate Le Cam's method with: H_0: $s(c; \mathcal{H}) \sim P_0 = \mathcal{N}(\mu_0, \sigma^2)$ (benign); H_1: $s(c; \mathcal{H}) \sim P_1 = \mathcal{N}(\mu_1, \sigma^2)$ (poisoned), $\mu_1 > \mu_0$. Then $\mathrm{KL}(P_0 \| P_1) = (\mu_1 - \mu_0)^2/(2\sigma^2) = \rho^2/2$.

Application. For product distributions, $\mathrm{KL}(P_0^N \| P_1^N) = N\rho^2/2$. By Pinsker: $\mathrm{TV}(P_0^N, P_1^N) \leq \sqrt{N\rho^2/4} = \rho\sqrt{N}/2$. Le Cam gives minimax risk $R^* \geq \frac{1}{2}(1 - \mathrm{TV}) \geq \frac{1}{2}(1 - \rho\sqrt{N}/2)$. Total error $\geq 2R^* = 1 - \rho\sqrt{N}/2$.

A.3 Proof of Theorem 11

By Hoeffding on $[0, 1]$: $|\hat{\mu}_N - \mu| \leq \sqrt{\log(2/\delta)/(2N)}$ w.p. $\geq 1 - \delta/2$. By chi-squared concentration: $|\hat{\sigma}_N - \sigma| \leq \sigma\sqrt{\log(2/\delta)/(2(N-1))}$ w.p. $\geq 1 - \delta/2$. Triangle inequality: $|\tau_N - \tau^*| \leq |\hat{\mu}_N - \mu| + \kappa|\hat{\sigma}_N - \sigma|$. Union bound yields Eq. (12).

A.4 Proof of Theorem 12

Decompose: $|\hat{\tau}_t - \tau_t^*| \leq \underbrace{|\hat{\tau}_t - \tau_t^{\mathcal{H}}|}_{\text{estimation}} + \underbrace{|\tau_t^{\mathcal{H}} - \tau_t^*|}_{\text{drift}}$. Estimation error: $O(\sigma(1 + \kappa)/\sqrt{m_{\max}})$ by Theorem 11. Drift error: cosine similarity is 1-Lipschitz on $\mathbb{S}^{d-1}$, so $|\mu_t^{\mathcal{H}} - \mu_t| \leq m_{\max}\Delta$. Setting equal: $\sigma/\sqrt{m_{\max}} = m_{\max}\Delta$ gives $m_{\max}^* = (\sigma/\Delta)^{2/3}$, yielding per-step regret $O(\sigma^{2/3}\Delta^{1/3})$.

B Additional defense details

Watermark. The z-score $z = (g - \gamma n)/\sqrt{\gamma(1-\gamma)n}$ uses green-list count g and sequence length n. Character-level: 95 ASCII chars, $\gamma = 0.45$. Token-level (GPT-2, 50,257 tokens): $p_{\text{green}} \approx 0.711$, mean $z \approx 7.5$ at $n = 50$.

Proactive. 16 probe queries span 8 subtopics. Threshold $\tau = 0.19$ minimizes FPR while maintaining ≥ 0.50 TPR on centroid passages.

Evaluation correction. Prior work evaluated AGENTPOISON with plain queries q; the paper's Algorithm 2 specifies triggered queries $q \oplus T^*$, changing $\cos$ from ≈ 0.45 to ~ 0.78.

C Ablation studies

D Multi-encoder evaluation

Key findings: (1) Triggered calibration achieves TPR $= 1.000$ for AGENTPOISON on all encoders, confirming Theorem 6 transfers across architectures. (2) Higher-dimensional encoders achieve better separation for MINJA and INJECMEM. (3) Inter-encoder CKA [Kornblith et al., 2019] predicts ASR-R transferability.

E Graph memory attacks

Definition 14 (Graph memory system)**.** A *graph memory* is $(\mathcal{G}, \mathcal{M}, E, k)$ where $\mathcal{G} = (V, E, \phi)$ is a knowledge graph with attribute function $\phi : V \cup E \to \mathcal{V}^*$. Retrieval combines em-

Table 5: MEMSAD threshold sensitivity with combined scoring.

	TPR			FPR		
κ	AP	MJ	IM	AP	MJ	IM
1.0	1.00†	1.00	0.80	0.10	0.10	0.10
1.5	1.00†	1.00	0.60	0.00	0.05	0.05
2.0	1.00†	1.00	0.40	0.00	0.00	0.00
2.5	1.00†	1.00	0.20	0.00	0.00	0.00
3.0	0.80†	0.80	0.00	0.00	0.00	0.00

†Triggered calibration. Plain: TPR $= 0.00$ for all κ.

Table 6: Corpus size ablation: ASR-R across $|\mathcal{M}|$ (triggered for AP).

$\lvert\mathcal{M}\rvert$	AGENTPOISON	MINJA	INJECMEM
50	1.00	0.70	0.65
100	1.00	0.65	0.55
200	1.00	0.65	0.50
300	1.00	0.60	0.40
500	1.00	0.60	0.30

bedding similarity with BFS traversal: $\mathrm{Ret}_{\mathcal{G}}(q,k,h) = \{v \in V : d_{\mathcal{G}}(v,v^*) \leq h,\ v^* \in \arg\max_{v'} \cos(E(q), E(\phi(v')))\}$.

Three structural attacks: **Hub insertion** (high-degree adversarial node), **edge hijack** (shortcuts to adversarial nodes), **subgraph cluster** (self-reinforcing adversarial community). Degree-anomaly detection flags $z_{\mathrm{deg}}(v) > \kappa_{\mathrm{deg}}$; adjacency contamination scoring detects edge-hijack via the fraction of recently added edges.

F Multi-agent propagation

Definition 15 (SIR propagation)**.** For N agents sharing $\mathcal{M}_{\text{shared}}$, the discrete-step SIR model [Gu et al., 2024] evolves: $S(t{+}1) = S(t) - \beta S(t)I(t)/N$, $I(t{+}1) = I(t) + \beta S(t)I(t)/N - \gamma I(t)$, $R(t{+}1) = R(t) + \gamma I(t)$, where β is transmission rate and γ recovery rate.

Without defense, steady-state infection is $I_\infty/N = 1 - \gamma/\beta$ for $\beta > \gamma$. With MEMSAD quarantine (TPR $= 1.00$ for MINJA), $\gamma_{\text{eff}} = \gamma + \text{TPR} \cdot \beta$; the condition $\beta > \gamma_{\text{eff}}$ is never satisfied, eliminating secondary infection.

G FPR validation

H Computational complexity

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: All five contributions listed in Section 1 are supported by theoretical results (Theorems 6–12, Lemma 8, Proposition 13) and experimental validation in Section 6.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

Table 7: Poison count ablation: ASR-R vs. n_{base}.

n_{base}	AGENTPOISON	MINJA	INJECMEM
1	0.80	0.20	0.10
3	1.00	0.45	0.30
5	1.00	0.65	0.50
7	1.00	0.70	0.55
10	1.00	0.75	0.60

Table 8: Encoder generalization: MEMSAD across 7 encoders ($\kappa = 2.0$, combined scoring).

	TPR			AUROC		
Encoder	AP	MJ	IM	AP	MJ	IM
MiniLM-L6 (d=384)	1.00	1.00	0.40	1.000	1.000	0.920
MPNet-Base (d=768)	1.00	1.00	0.40	1.000	1.000	0.900
Para-MiniLM (d=384)	1.00	0.80	0.20	1.000	0.960	0.840
E5-Base (d=768)	1.00	0.80	0.40	1.000	0.940	0.880
Contriever (d=768)	1.00	1.00	0.40	1.000	1.000	0.900
BGE-Large (d=1024)	1.00	1.00	0.60	1.000	1.000	0.940
OpenAI-small (d=1536)	1.00	1.00	0.60	1.000	1.000	0.960

FPR $= 0.00$ for all encoders at $\kappa = 2.0$. AP uses triggered calibration.

Justification: Section 7 includes a dedicated limitations paragraph covering corpus size, modelled ASR-A, query distribution stationarity, the synonym loophole, and single-encoder experiments.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: Theorem 6 states all assumptions (Assumption 4) with proof in main text and geometric interpretation in Appendix A.1. Lemma 8 provides a certified detection radius with complete proof. Theorem 9 has proof sketch in main text and full proof in Appendix A.2. Theorems 11 and 12 have full proofs in Appendices A.3 and A.4. Proposition 13 has a complete proof in the main text.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Section 6 specifies the embedding model (all-MiniLM-L6-v2, 384-dim), index type (FAISS IndexFlatIP), corpus size (200), poison counts, top-k, and number of seeds (5). The full pipeline is reproducible via a single script.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: All code, the 200-entry synthetic corpus, evaluation scripts, and figure generation scripts are open-sourced. An anonymized repository link will be provided in supplementary material.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Table 9: FPR validation: Clopper-Pearson 95% CIs, 20 trials.

Defense	FPR	95% CI	n/trial
MEMSAD ($\kappa = 2.0$)	0.000	$[0.000, 0.004]$	1000
Watermark ($z_{\text{thr}} = 4.0$)	0.000	$[0.000, 0.004]$	1000
Proactive ($\tau = 0.19$)	0.010	$[0.004, 0.023]$	1000

Table 10: Per-entry defense cost at write time.

Defense	Time	Space	Measured
MEMSAD (exact)	$O(md)$	$O(md)$	0.8ms
MEMSAD (ANN)	$O(d \log m)$	$O(md)$	0.3ms
Watermark	$O(\|c\| \cdot \|\mathcal{V}\|)$	$O(\|\mathcal{V}\|)$	1.2ms
Validation	$O(\|c\| \cdot \|P\|)$	$O(\|P\|)$	0.1ms
Proactive	$O(\|Q_p\|d)$	$O(\|Q_p\|d)$	0.5ms
Composite	$\max_j O(d_j)$	$\sum_j O(s_j)$	2.1ms

Apple M-series, 16GB; $|\mathcal{M}| = 200$, $m = 20$, $d = 384$.

Answer: [Yes]

Justification: Section 6 and Appendix C specify all hyperparameters: $\kappa = 2.0$ for MEMSAD, $z_{\text{thr}} = 4.0$ for watermark, $\tau = 0.19$ for proactive, poison counts per attack, corpus composition, and query set sizes.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Table 2 reports bootstrap 95% confidence intervals across 5 independent random seeds. Appendix G validates FPR over 20 independent trials using Clopper-Pearson exact binomial CIs.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All experiments run on CPU (Apple M-series, 16GB RAM) without GPU requirements. Appendix H reports per-entry timing for each defense. The full pipeline completes in under 5 minutes.

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Justification: All models (all-MiniLM-L6-v2, GPT-2, DPR) and libraries (FAISS, sentence-transformers) are properly cited under permissive licenses (Apache 2.0, MIT).

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

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Justification: The MEMSAD framework, synthetic corpus, and evaluation pipeline are documented with API documentation and usage examples.

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Answer: [Yes]

Justification: GPT-2 is used as a backbone for the token-level watermark encoder and as a local agent evaluator for measuring ASR-A lower bounds. These uses are described in Section 5 and Appendix B.